

Data Validation & Consistency

Checks Log — HeavySuppliersWarehouseDatasets

Checks that PASSED (numbers reconcile, no action taken)

Check	Result
sum(sales_orders_lines.line_grand_total) per SO == sales_orders_header.grand_total	Match on all 20,000 orders (float noise only)
sum(purchase_orders_lines.line_grand_total) per PO == purchase_orders_header.grand_total	Match on all 24,000 POs
invoices.grand_total == linked sales_orders_header.grand_total	Match on all 18,033 invoices
line_total x gst_rate/100 == gst_amount (sales lines)	Match on all 130,402 lines
line_total x gst_rate/100 == gst_amount (purchase lines)	Match on all 155,495 lines
line_total + gst_amount == line_grand_total (sales & purchase lines)	Match on all rows, both tables
sum(payments) per invoice <= invoice grand_total	0 overpayments (checked on the 18,840 unambiguous payments only - see cleaning log for the 417 ambiguous ones)
customers.current_balance <= customers.credit_limit	0 customers over limit, all 500 checked

Checks that FAILED — fixed or flagged below

1. stock_ledger: ADJUSTMENT rows have an unreliable quantity sign

Reconstructing running_balance step-by-step from signed quantities (IN = +, OUT = -) matched perfectly for all IN/OUT rows, but 1,222 of 2,454 ADJUSTMENT rows didn't: quantity is stored as a positive number even when the balance actually decreased. Tested the hypothesis that ADJUSTMENT is always a decrease — still 1,219 mismatches, so there's no simple sign rule to apply. - **Fix:** added implied_signed_quantity = the actual balance delta (running_balance minus the previous row's running_balance for that product/branch) — this is reliable for all 237,230 rows regardless of movement_type. Added adjustment_sign_ambiguous_flag (True for all 2,454 ADJUSTMENT rows) as a signal to use implied_signed_quantity instead of the raw quantity column for any signed-quantity analysis.

2. products: one margin_percentage doesn't match its own cost/price

P002 (Engine Oil Filter OF-90): unit_cost=450, unit_price=720 implies a 60.0% margin, but margin_percentage is stored as 37.3%. All 29 other products check out exactly. - **Fix:** added calculated_margin_percentage and margin_mismatch_flag to product_features.csv. Did NOT overwrite the stored value — a 37.3% margin could reflect an earlier cost basis we don't have visibility into, so this needs a human call, not a silent overwrite.

Files changed in this stage

- stock_ledger.csv: +2 columns (implied_signed_quantity, adjustment_sign_ambiguous_flag)
- product_features.csv: +2 columns (calculated_margin_percentage, margin_mismatch_flag)
- All other files carried through unchanged from the feature-engineering stage.